MODULE 5

Sea Wave Energy & Ocean Thermal Energy

INTRODUCTION

Broadly the ocean sources of energy are Ocean Thermal Energy Conversion (OTEC) and the Tidal energy, Wave energy and fourth form of the energy that emanates from the sun-ocean system stems from the mechanism of surface water evaporation by solar heating i.e. hydrological cycle.

- These energy sources (except tidal) are the result of the absorption by the seas and oceans of solar radiation, which causes, like the wind, ocean currents and moderate temperature gradients from the water surface downward, especially in tropical waters. The oceans and seas constitute some 70 per cent of the earth's surface area, so they represent a rather large storage reservoir of the solar input.
- The conversion of solar energy stored as heat in the ocean into electrical energy by making use of the temperature difference between the warm surface water and the colder deep water. The facilities proposed for achieving this conversion are commonly referred to as OTEC plants or sometimes as solar sea power plants (SSPP).
- Since the ocean waters are heated by the sun, they constitute a virtually in- exhaustible source of energy. However, unlike direct solar energy, the ocean energy is available continuously rather than only in the daytime.
- The operation of the OTEC plant is based on a well-established physical (thermodynamic) principle. If a heat source, is available at a higher temperature and a heat sink at a lower temperature, it is possible in principle, to utilize the temperature difference in a machine or prime mover (e.g. a turbine) that can convert part of the heat taken up from the source into mechanical energy and hence into electrical energy. The residual heat is discharged to the sink at the lower temperature.
- In the OTEC system, the warm ocean surface water is the heat source and the deep colder water provides the sink.
- The temperature gradient can be utilized in a heat engine to generate power. This is called ocean thermal energy conversion (OTEC).

Ocean Thermal Electric Conversion (OTEC)

The ocean thermal energy concept was proposed as early as 1881 by the French physicist Jacques d' Arsonval. In this indirect form of solar energy at sea, collection and storage are free. The surface of the water acts as the collector for solar heat while the upper layer of the sea constitutes infinite heat storage reservoir.

Thus heat contained in the oceans, which is solar in origin could be converted into electricity by utilizing the fact that the temperature difference between the warm surface water of the tropical oceans and the colder waters in the depths is about 20-25°K.

Warm surface water could be used to heat some low boiling organic fluid, the vapour of which would run a heat engine. The exit vapour would condensed by pumping cold water from the deeper regions. The amount of energy available for ocean thermal power generation is enormous, and is replenished continuously.

Solar energy absorption by the water takes place according to Lambert's law of absorption, which states that each layer of equal thickness absorbs the same fraction of light that passes through it.

Mathematical

$$-\frac{dI_{(x)}}{dx} = kI$$

$$\text{Or} I_{(x)} = I_o e^{-kx}$$

Where Io and I(x) are the intensities of radiation at the surface (x = 0) and at a distance x below the surface. K is an extinction coefficient (or absorption coefficient) that has the unit $\mathrm{L^{-1}}$, K has values of

0.05 $\mathrm{m^{-1}}$for very clear fresh water, 0.27 for turbid fresh water and 0.50 $\mathrm{m^{-1}}$for very salty water.

Thus the intensity decreases exponentially with depth and, depending upon K, almost all of the absorption occurs very close to the surface of deep waters.

Because of heat and mass transfer at the surface itself, the maximum temperatures occur just below the surface. Considering deep water in general, the high temperatures are at the surface, whereas deed water remains cool. In the tropics, the ocean surface temperature often exceeds 25°C, while 1 km below, the temperature is usually not higher than 10°C. Water density decreases with an increase in temperature (above 4°C where pure water's density is maximum, decreasing again below this temperature, the reason ice floats).

Thus there will be no thermal convection currents between the warmer, lighter water at the top and deep cooler, heavier water. Thermal conduction heat transfer between them across the large depths is too low to after this picture, and thus mixing the retarded, so the warm water stays at the top and the cool water stays at the bottom. It is said, therefore, that in tropical waters there are two essentially infinite heat reservoirs, a heat source at the surface at about 27°C and a heat sink, some 1 km directly below, at about 4°C; both reservoirs are maintained annually by solar incidence. The concept of ocean thermal energy conversion (OTEC) is based on the utilization of this temperature difference in a heat engine to generate power.

The surface temperatures (and temperature differences) vary both with latitude and season, both being maximum in tropical, subtropical, and equatorial waters i.e., between the two tropics, making these waters the most suitable for OTEC systems. Several such plants are built in France after World War II (the largest of which has a capacity of 7.5 MW).

With a 22°K temperature difference between surface and depths, such as exists in warmer ocean areas than in North Sea, the Carnot efficiency is around 7%. This is obviously very low, and comparable to that expected from a flat plate collector.

In fact, by the time the overall efficiency has been reduced by using a practical engine (operating on a Rankine cycle say) together with heat exchangers, the propositions might seem hopeless. One major difference between these two heat sources is that solar energy arrives with a 10W power density, and requires a large acreage of flat-plate collector.

Whereas on ocean thermal gradient source can operate with a small area collector by pumping sufficient water through the heat collector. Indeed the attraction of the solar sea power plant lies in its present day engineering feasibility and possible competitive cost with fossil fuel power stations.

As stated the idea of ocean thermal energy conversion with a suitable working fluid was originated by d' Arsonval, but the technical feasibility of the open cycle system was demonstrated by Claude with an installation on the south coast of Cuba in 1929.

It was a remarkable achievement at the time. The electric power generated was 22 kilowatts with an overall efficiency more than 1 per cent. The hot and cold water were conducted through the long pipes to the machinery ashore. With the limited technology and cheap fuel at that time, there was then little prospects for economic feasibility.

A larger installation with two units totalling 7 megawatts was constructed on the Ivory coast by the French in 1956, but encountered troubles and was abandoned.

The process of OTEC, requires that the warm surface water and cold water from depth (about 1000-1500 m), be brought into proximity so they act as the heat source and the heat sink, respectively for a heat engine.

In other words, solar energy collected and stored as heat by the world's major oceans, can be converted into electricity through a generation process similar to that of conventional power plants, except that in the case of OTEC, no depletable fuel is required.

Furthermore, although there is some seasonal variation in the ocean thermal resource at a given OTEC power plant location, there is little diurnal variation. Accordingly OTEC power plants are analogous to solar hydropower plants in that they smooth out the diurnal intermittence of the solar radiation, in contrast to other electric power options.

OTEC power plants provide a potentially substantial renewable source of base load electricity, .albeit located mainly at sea. Although it is possible to find good land sites where OTEC power plants can be located, by bringing the warm and cold water onto shore via aqueducts (artificial cannal/conduit), it is clear that such opportunities will be much limited on a global basis than the ample opportunities for generating substantial amounts of OTEC electricity abroad floating OTEC platforms. This is both because of the special technical requirements for on shore OTEC plants and because of the limited market potential (at least in the near term) for OTEC electricity at such sites.(n shore

OTEC power plants will be viable mainly at locations where three requirements are all simultaneously satisfied with satisfactory economics:

(i) Coastal zone land must be available,

(ii) Sea floor must descend sufficiently rapidly from the shore based plant location; and

The seasonal availability of warm and cold water without undue gradation by the war and cold water effluents from the OTEC plant must meet certain criteria. In any event, it is probable that available and attractive on shore and near shore OTEC power plant location will be populated early in the development and implementation of the OTEC concept, both as convenient locations for pilot and demonstration plants and because they will constitute attractive intermediate markets for OTEC electricity and by products.

OTEC power generation system gives less efficiency, as stated above. However, because of the OTEC requirement for paas1tic power (such as for pumping up the cold water supply) and other losses, the achievable net conversion efficiency is only about 2.5 percent (Carnot efficiency 7'). This compares a net efficiencies of 30 to 40% associated with conventional power

Some engineers question whether such as an extremely low net efficiency will ever allow OTEC to become economically viable. However, it is important to consider the matter in more sophisticated terms than net efficiency; since in the case of OTEC there is no fuel cost, only the requirement to pay for circulating much more warm and cold water, than is normally associated with power generation. This means that extensive areas heat exchangers will be required for "closed cycle" OTEC plants (which would employ a working fluid such as ammonia) or those degasifiers (to remove gases dissolved in the sea water) and tremendous turbines would be required for "open cycle" OTEC plants that would operate by the flash evaporation of sea water.

Thus, although the net efficiency of the JEC plant must certainly be positive and as high as is readily attainable, the key economics question is the resulting cost of OTEC electrical energy, not the actual value of the net efficiency.

METHODS OF OCEAN THERMAL ELECTRIC POWER GENERATION:

There are two rather different methods for harnessing ocean thermal differences. One is the open cycle*,* also known as the Claude cycle, and other is the *closed cycle* system, also known as the Anderson cycle.

In the closed cycle system**,** a liquid working fluid, such as ammonia or propane, is vaporized in an evaporator (or boiler); the heat required for vaporization is transferred from the warm ocean surface to the liquid by means of a heat exchanger'*)*

The high-pressure vapour leaving the evaporator drives an expansion turbine, similar to a steam turbine that it is designed to operate at a lower inlet pressure. The turbine is connected to an electric generator in the usual manner.

The low pressure exhaust from the turbine is cooled and converted back into liquid in the condenser. The cooling is achieved by passing cold, deep ocean water, from a depth of 700 to 90O or more, through a heat exchanger. T

The liquid working fluid is then pumped back as high pressure liquid to the evaporator, thus closing the In the open-cycle turbine system, water is the working fluid. The warm surface water is caused to boil by lowering the pressure, without supplying any additional heather low-pressure steam produced then drives a turbine, and the exhaust steam is condensed by the deep colder water and is discarded.

A heat exchanger is not required in the evaporator and direct-contact between the exhaust steam and a cold- water spray makes a heat exchanger as necessary in the condenser.

On the other hand, because of the low energy content of the low pressure steam, very large turbines or several smaller units operating in parallel would be required to achieve a useful electric power output.

The Claude cycle or open cycle which is older one, utilizes the vapour pressure of sea water itself as the working medium and has been demonstrated to be practicable.

The other method, a closed cycle known as the Rankine cycle, uses a working fluid with higher vapour pressure (such as ammonia, hydrocarbon or halocarbon) at the temperature available.

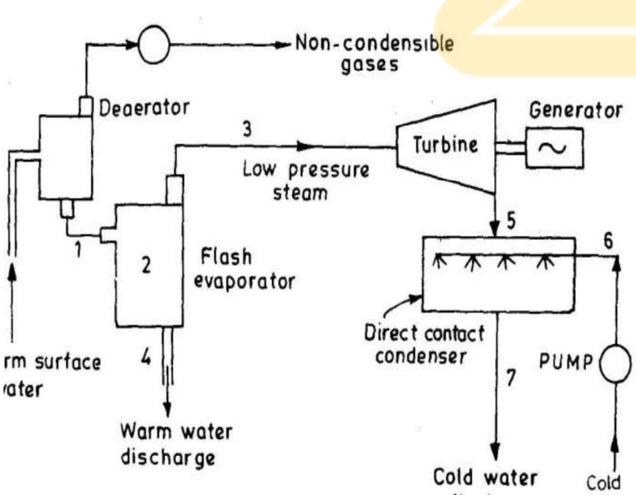
This cycle is favoured for the future development in expectation of higher efficiency.

OPEN CYCLE OTEC SYSTEM (CLAUDE CYCLE)

'Open cycle' refers to the 'utilization of sea water as the working fluid, wherein sea water is flash evaporated under a partial vacuum. The low pressure steam is passed through a turbine, which extracts energy from it, and then the spent vapour is cooled in a condenser.

This cycle drives the name 'open' from the fact that the condensate need not be returned to the evaporator, as in the case of the 'closed cycle'.

Instead, the condensate, can be utilized as desalinated water if a surface condenser is used, or if a spray (direct-contact) condenser is used, the condensate is mixed with the cooling water and the mixture is discharged back into the ocean. &schematic diagram of the open cycle system is shown in Fig.



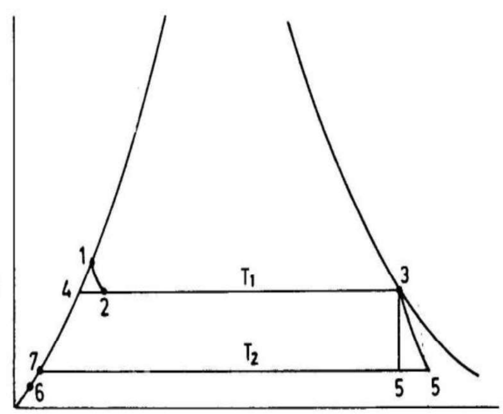


Fig.Schematic of the OTEC open cycle. **Fig. T.S. diagram corresponding to Fig.**

Its corresponding T-S diagram is also Fig. Schematic of the OTEC open cycle. Shown in the Fig. In the cycle shown warm surface water at say 27°C is admitted into an evaporator in which the pressure is maintained at a value slightly below the saturation pressure

At the new pressure, water which is entering the evaporator gets 'super-heated'. As shown in Fig. the warm water which is at 27°C, has a saturation pressure of 0.03619 kg/cm2 (0.0356 bar) (point 1).

The evaporator pressure is 0.03213 (0.0317 bar), which corresponds to 25°C saturation temperature. This temporarily superheated water undergoes *volume boiling* (as opposed to pool boiling which takes place in conventional boilers due to an immersed heating surface), causing that water to partially flash to steam to an equilibrium two phase condition at the new pressure and temperature of 0.032 13 kg/km2 and 25°C (point 2). Process 1-2 is a throttling and hence constant enthalpy process.

The low pressure in the evaporator is maintained by a vacuum pump that also removes the dissolved non-condensable gases from the evaporator.

At point 2, the evaporator contains a mixture of water and steam of very low quality. The steam is separated from the water as saturated vapour at 3.

The remaining water is saturated at 4 and is discharged as brine back to the ocean. The steam at 3, has a very low pressure and high specific volume (0.03213 kg/cm2,43.40 m3/kg), as compared to conventional fossil power plant, which has about 160 kg/cm2 pressure and 0.021 m3/kg specific volume.

The steam expands in a specially designed turbine that can handle such conditions to 5. The condenser pressure and temperature at point 5 are of the order of 0.01729 kg/cm2 (0.0 17 bar) and 15°C. A direct contact condenser is used as the turbine exhaust steam will be discharged back to the ocean in the open cycle system. In the condenser, the exhaust steam is mixed with cold water from the deep cold water pipe at 6, which results in near saturated water at 7. This water is allowed to discharge to the ocean.

The cooling water from the deep ocean which is at about 11°C, on reaching the condenser, its temperature rises to about 15°C, due to heat transfer between the progressively warmer outside water and cooling water inside the pipe, as it ascends towards the top.

It can be seen that very large ocean water mass and volume flow rates are used in open OTEC systems and that the turbine is a very low pressure until that receives steam with specific volumes more than 2000 times that in a modern fossil power plant. Thus the turbine resembles the few lost exhaust stages of a conventional turbine and is thus physically large. Because of the need in the open cycle to harness the energy in low pressure steam, extremely large turbines (compared to wind turbines) must be utilized. Further-more degasifiers (deaerators) must be used to remove the gases dissolved in the sea water unless one is willing to accept large losses in efficiency.

On the other hand, since there are no heat transfer problems in the evaporator, the problem of bio-fouling control is minimized. The cost of an open-cycle system for providing substantial number of megawatts is presently regarded by most OTEC workers as being significantly greater than for closed cycle system. The turbine cost constituted almost half the cost of the power system, but may be amenable to reductions that could result from design innovations.

The Closed or Anderson, OTEC Cycle

A schematic of a closed-cycle OTEC power plant is shown in Fig. Heat exchanger known as evaporators and condensers are a key ingredient, since extensive areas of material are needed to transfer significant amounts of low quality heat of the low temperature differences being exploited.

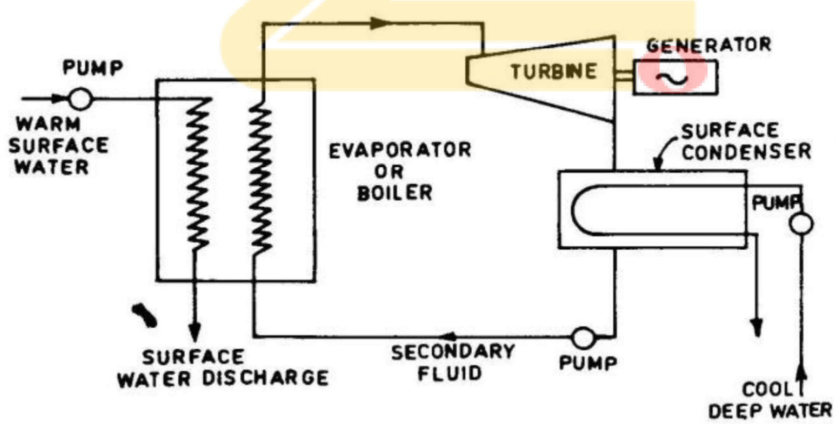


Fig. Schematic of an OTEC closed cycle system.

In other words, large volumes of water must be circulated through the OTEC power plant, requiring commensurately large heat exchangers. The actual components employed in an OTEC closed cycle system would appear more like the hardware illustrated in Fig another closed cycle schematic. This cycle requires a separate working fluid that receives and rejects heat to the source and sink*via* heat exchangers (boiler or evaporator and surface condenser). The working fluid may be ammonia, propane, or a Freon.

The operating of such fluids at the boiler and condenser temperatures are much higher than those of water, being roughly10 kg/cm2 (10 bar) at the boiler, and their specific volumes are much lower, being comparable to those of steam in conventional power plants.

Such pressures and specific volumes result in turbines that are much smaller and hence less costly than those that use the low pressure steam of the open cycle. The closed cycle also avoids the problems of the evaporator. It however, requires the use of very large heat exchangers (boiler and condenser) because, for an efficiency of about 2 percent, the amounts of heat added and rejected are about 50 times the output of the plant. In addition, the temperature differences in the boiler and condenser must be kept as low as possible to allow for the maximum possible temperature difference across the turbine, which also contributes to the large surfaces of these units.

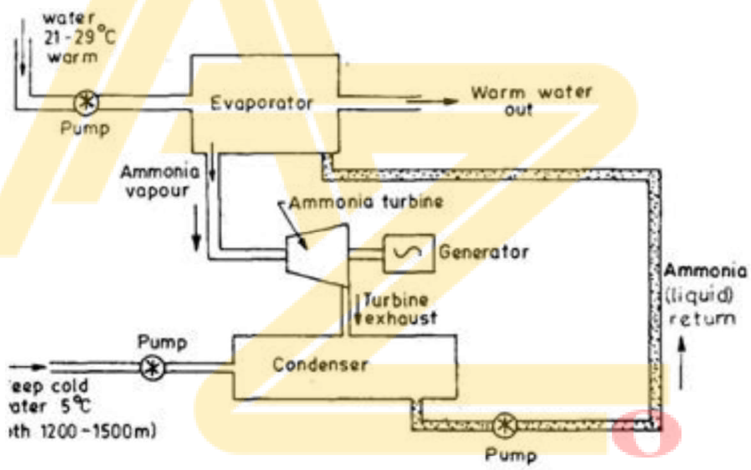


Fig. 2.Schematic of a closed OTEC ammonia cycle.

The closed cycle approach was first proposed by Barjot in 1926, but the most recent design was by Anderson and Anderson *in* the 1960s. The closed cycle is sometimes referred to as the Anderson Cycle. In the cycle propane was chosen as the working fluid. The temperature difference between warm surface and cool surface was 20°C. The c surface was at about 600 m deep. Propane is vaporized in the boilei evaporator at about 10 kg/cm2 (10 bar) or more and exhausted in 1 condenser at about 5 bar. Instead of usual heavier and more expensive shell and tube h exchangers, the Anderson OTEC system employs thin plate type h exchangers, which minimizes the mass and the amount of material a hence cost. The heat exchangers are placed at depths where the sta pressure of the water in either heat exchanger roughly equals i pressure of the working fluid, this helps in reducing the thickness of plates. A fundamental requirement in closed cycle systems is to trans. heat efficiently across the heat exchanger surfaces constituting t evaporators and condensers, so as to achieve a high value of overall h transfer coefficient (U) measured in watts per kelvin per square meter or W/°K/m2. For the evaporation, this overall heat transfer coefficient is a measure of how effectively heat is transferred sequentially from s water through the heat exchanger materially (a metallic alloy) a hence to the working fluid *(e.g.,* ammonia). For the condenser, an over U characterized the reverse heat transfer process.

In an ocean environment, it is likely that a layer of slime known as "bio fouling" will *eventually* accumulate on the water side of the he exchangers. Such slime is first comprised or micro-organisms, at which stage, the bio fouling is called "micro fouling". Subsequently, if the slime is not removed, additional bio-fouling in the form of micro-organism will become attached, augmenting the slime layer. The occurrence micro-fouling seems to be a pre-requisite for the attachment of macro organisms. A film of corrosion and possibly of calcareous *(e.g.* minerals deposits can also accumulate on the water side (and conceivably through leakage—even on the working fluid side of the heat trans surfaces). The total formulation of bio-fouling, corrosion, and so on, referred to a "fouling" (or scaling) and will tend to inhibit heat trans through it. The "fouling factor" is a measure of the thermal resistance'R_f'of a fouling film. This thermal resistance is the reciprocal of t corresponding heat transfer coefficient 'h_f'of the fouling film.

To maintain viable OTEC heat exchangers, provisions must be made to inhibit the formation of fouling layers and to remove any significant fouling that forms. Removal can be accomplished by periodically cleaning ti heat exchanger surfaces through mechanical, chemical or other means.

Although both closed-and open cycle turbine systems are being explored, it appears that closed-cycle systems offer the most promise for the near future. Each of the possible working fluids *(i.e.* ammon and propane) has advantages and disadvantages.

SITE SELECTION

- In selecting a site for an OTEC facility, the primary consideration is, of course, a significant temperature difference—at least about 20°C—between surface and deep ocean waters (for 700-900 m depth or more) that will permit year round operation.
- The greater the difference, the lower will be the cost of generating electricity. The best sites are in the tropical belt between about 20°N and 20°S latitude. There are, however, several locations outside this area that might be suitable for OTEC plants. In choosing a site, consideration should be given to the potential for bio-fouling effects as noted earlier.
- As a general rule, an OTEC plant would be located offshore in order to provide access to the deep colder water. However, an ideal situation might be one where the shoreline dropped steeply to a considerable depth. Most of the installation could then be more conveniently build on land.

ENERGY UTILIZATION

If possible an OTEC plant should be less than about 30 km from shore. The electricity generated could then be transmitted inexpensively to land by submarine cable. At somewhat greater distances, the trails- mission costs would be increased but might be tolerable.

If the plant is so far from shore that these costs become prohibitive, the electricity generated can be utilized at the plant site to produce energy-intensive materials.

One suggestion is to use direct electric current to decompose sea water by the process of electrolysis; the main products would be hydrogen and oxygen. The hydrogen could be liquified and transported by tanker to a point where it could be used as fuel. Alternatively, the hydrogen could be combined with atmospheric nitrogen to form ammonia for use as fertilizer, thereby saving natural gas which is presently the main source of hydrogen for this purpose.

HYBRID CYCLE

There are several variations on the standard OTEC open-cy system. One variation is the “hybrid cycle” which is an attempt combine the best features and avoid tlie worst features of the open closed cycles. First, as shown in Fig. 9.2.5.1, sea water is flash evaporated to steam, as in the open cycle. The heat in the resul steam is then transferred to ammonia in an otherwise conventional closed Rankine cycle system.

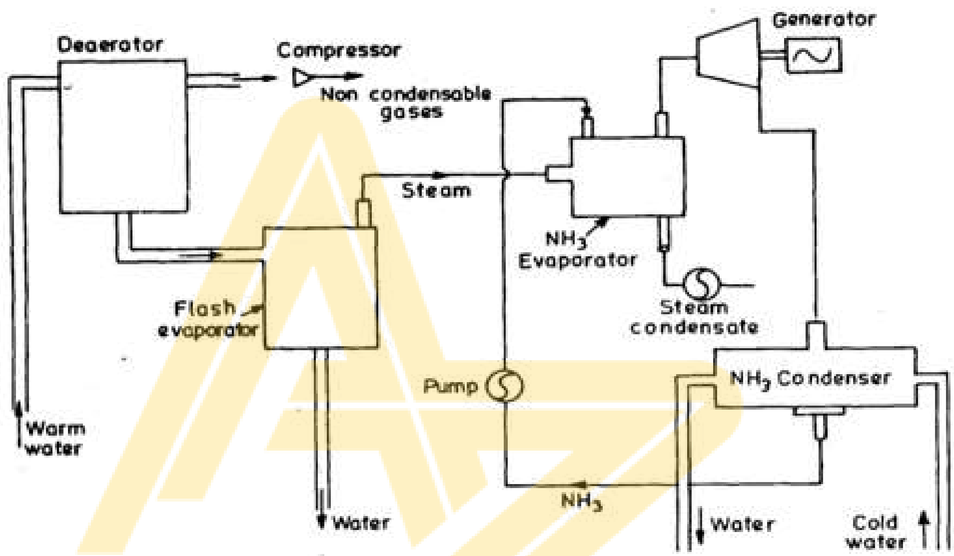


Fig. Hybrid Cycle

PROSPECTS OF OCEAN THERMAL ENERGY CONVERSION IN INDIA

The OTEC project cell established at IIT, Madras has completed the preliminary feasibility study for establishing a, 1 MW OTEC plant in Lakshadeep Island at Minicoy. The OTEC works on the principle of utilizing the temperature difference of sea water at depth and that at the surface.

The surface sea water is used to vaporize a low boiling chemical which drives, a turbo generator. The vaporized chemical is then compressed; it is condensed by using cold sea water from depth. Preliminary oceanographic studies on eastern side of Lakshadeep Island suggest the possibility of the establishment of shore based OTEC plant at the island with a cold water pipe line running down the slope to a depth of 800—1000 m.

Both the island has large lagoons on the western side. The lagoons are very shallow with hardly any nutrient in the sea water. The proposed OTEC plant will bring up the water from 1000 m depth which has high nutrient value. After providing cooling effect in the condenser, a part of deep sea water is proposed to be diverted to the lagoon for the development of aqua culture. A hydrographic survey of the proposed site was undertaken by National Hydrographic Office, Dehra Dun. The preliminary assessment of survey indicates the availability of suitable conditions for establishment of OTEC plant.